

German Electricity Demand Forecasting Using Statistical, Machine Learning and Deep Learning Models

Student Name: *Bhanu Prakash Mallam*

Student ID: 24095576

GitHub Repository Link: <https://github.com/bhanu7959/Time-Series-Modelling-Case-Study>

1. Introduction

Forecasting of electricity demand is an essential aspect of modern energy systems. Forecasts help electricity suppliers balance electricity supply with its demand and lower operating costs. Moreover, forecasting can contribute to the stability of energy systems through proper scheduling of operations. Forecasting of electricity demand can also be used for renewable energy management purposes.

Electricity consumption has complicated patterns over time that are affected by daily activities, weather during different seasons, economic performance, and public holidays. Such attributes make electricity consumption forecasting a difficult time series problem. Due to this, various forecasting techniques have been designed; they range from traditional statistics-based forecasting benchmarks to machine learning and deep learning algorithms.

This research aims to conduct an analysis on the accuracy of forecasting through various modeling techniques using the electricity demand in Germany. First, the data will be explored in order to gain insight into the nature of the time series. Then, benchmark forecasting methods, SARIMA, SARIMAX, Random Forest regression, and LSTM models will be applied to the time series.

Forecasting accuracy for each of the models is tested using MAE, RMSE, and sMAPE for the last two years of the available observations. After that, all models are compared to see if more complex models can outperform basic models in forecasting.

2. Dataset Description

The data set that was utilized in this study has been collected from the Open Power System Data (OPSD). This is a data source that makes electricity market data for some countries in Europe publicly accessible. Among the data sets, we have chosen the time series for electricity consumption in Germany (code DE).

The only observations kept in this analysis were those made from **1 January 2015 to 31 October 2020**. The initial hour-based dataset was converted to daily and weekly timeseries through resampling by Pandas. Weekly demand data were used for the benchmark, SARIMA,

SARIMAX, and Random Forest models, while the hourly observations were kept for the LSTM model.

In order to increase the accuracy of prediction, the historical temperature records of Berlin were obtained through Open-Meteo Historical Weather API in the form of weekly average temperatures. These were used as explanatory variables for the SARIMAX and Random Forest models.

The data cleaning process was carried out before carrying out the modelling process by transforming the time stamps to the datetime format, organizing observations in chronological order and dealing with any missing values through interpolation. Additionally, the data discrepancies in the time zones between the two data sets were sorted out.

3. Exploratory Data Analysis (EDA)

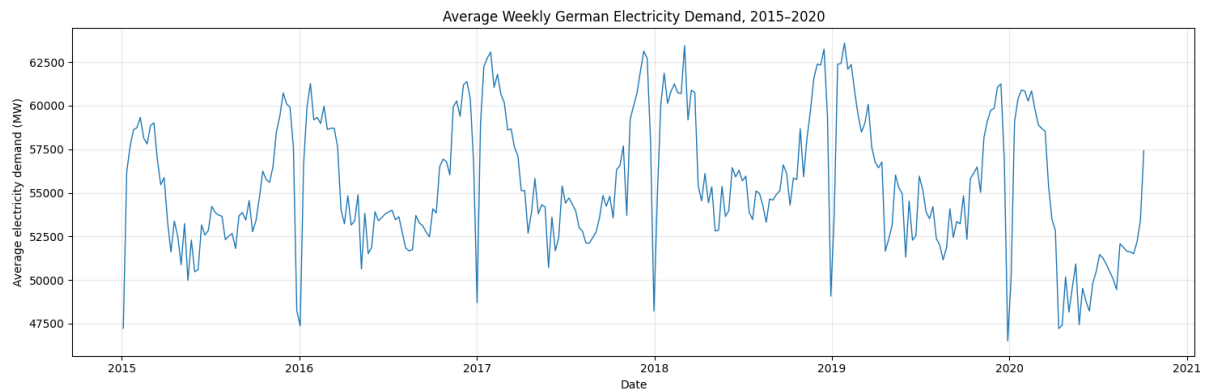
The exploratory data analysis (EDA) was performed to gain insight on the nature of the German electricity demand data prior to building models for forecasting. Weekly aggregate data for the period from January 2015 to October 2020 were used for statistical and machine learning models, whereas hourly data were used for LSTM modeling.

Table 1. Dataset Summary

Characteristic	Value
Country	Germany (DE)
Data Source	Open Power System Data (OPSD)
Study Period	January 2015 – October 2020
Original Frequency	Hourly
Forecast Frequency	Weekly (Parts 2–5), Hourly (Part 6)
Forecast Horizon	Final 104 Weeks

3.1 Weekly Electricity Demand

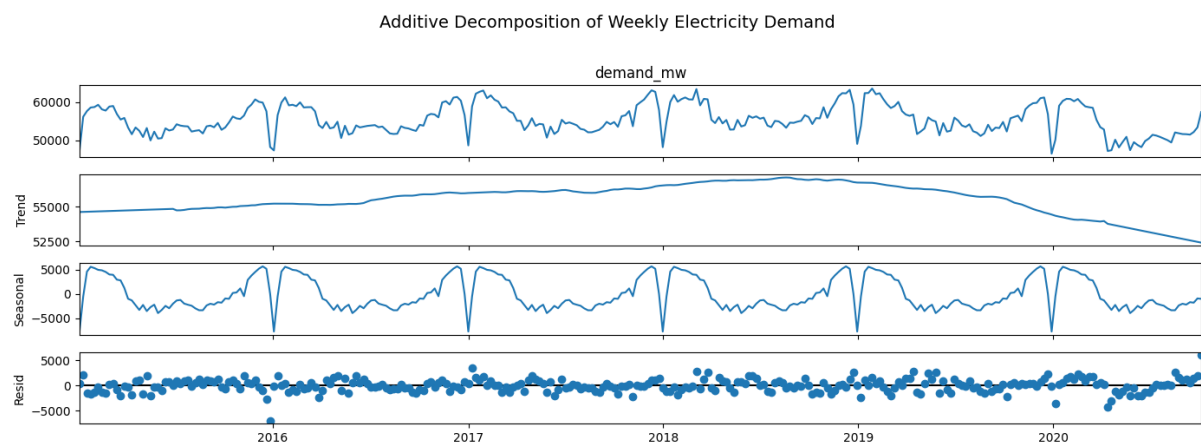
Figure 1. Weekly German Electricity Demand (2015–2020)



The yearly demand for electricity follows a seasonal trend with consistent fluctuations within the observation period. The demand increases when temperatures fall and decreases as the weather gets warm, meaning that seasonality is one of the key features of this dataset.

3.2 Seasonal Decomposition

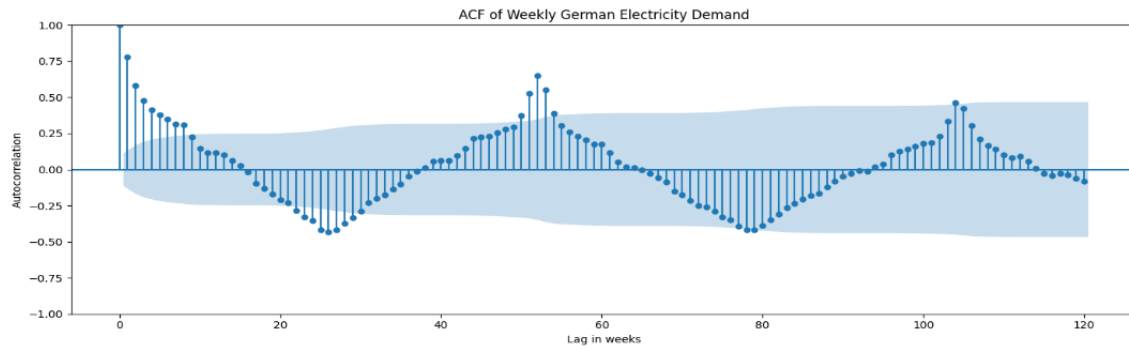
Figure 2. Seasonal Decomposition of Weekly Electricity Demand



Decomposition of the time series was done on seasonal basis, which resulted in the identification of three factors - trend, seasonality, and residuals. The decomposition revealed that there is significant and consistent annual seasonality, but the trend component was more or less stable over the entire time period of our analysis.

3.3 Stationarity Analysis

Figure 3. Autocorrelation Function (ACF) of Weekly Electricity Demand



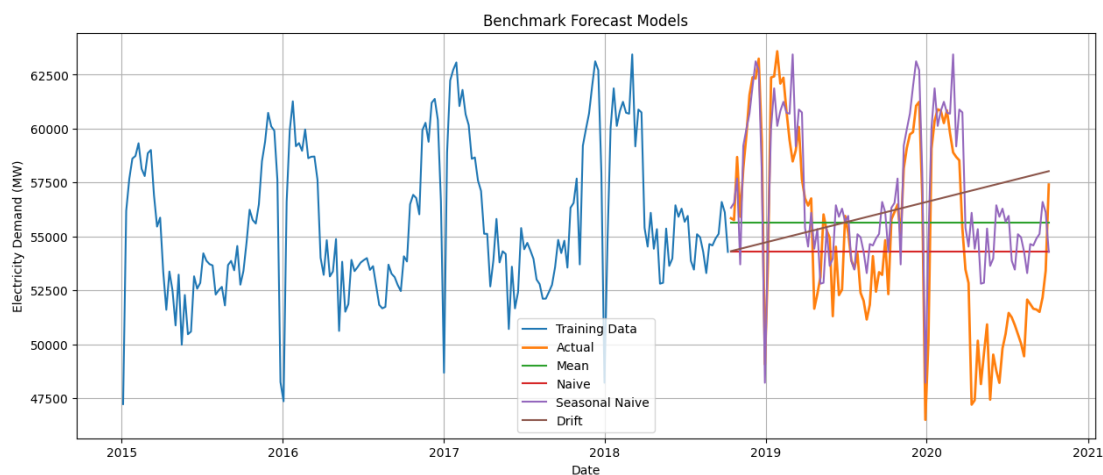
ACF plot revealed autocorrelation at various seasonal lags, which means that past demand observations can be used for future demand forecasting. The initial time series is non-stationary because of its seasonal nature; thus, the time series was differenced before the SARIMA model estimation.

In general, exploratory data analysis revealed that German electricity demand demonstrates strong seasonality on an annual basis. These properties can be considered the justification for using seasonal models and machine learning techniques for forecasting.

4. Benchmark Forecasting Models

These benchmarks were designed to serve as the benchmark models for comparison prior to using more sophisticated techniques. Four different forecasting models were considered on the basis of the weekly electricity demand data, including: Mean, Naïve, Seasonal Naïve and Drift models. The last 104 weeks of data were set aside for testing purposes, whereas the rest was utilized for training.

Figure 4. Benchmark Forecasts Compared with Actual Weekly Electricity Demand



The Mean and Naïve models did not account for the seasonality in electricity demand, leading to poor predictions. The Drift model captured the trend but could not replicate the annually recurrent seasonality. On the other hand, the Seasonal Naïve model was able to

follow the demand very closely since it simply replicated the demand from the same week of the previous year.

Forecast Evaluation

Table 2. Performance of Benchmark Forecasting Models

Model	MAE	RMSE	sMAPE (%)
Mean	3,788.83	4,397.30	6.87
Naïve	3,783.20	4,459.11	6.86
Seasonal Naïve	2,318.52	3,006.76	4.26
Drift	4,339.89	5,117.96	7.85

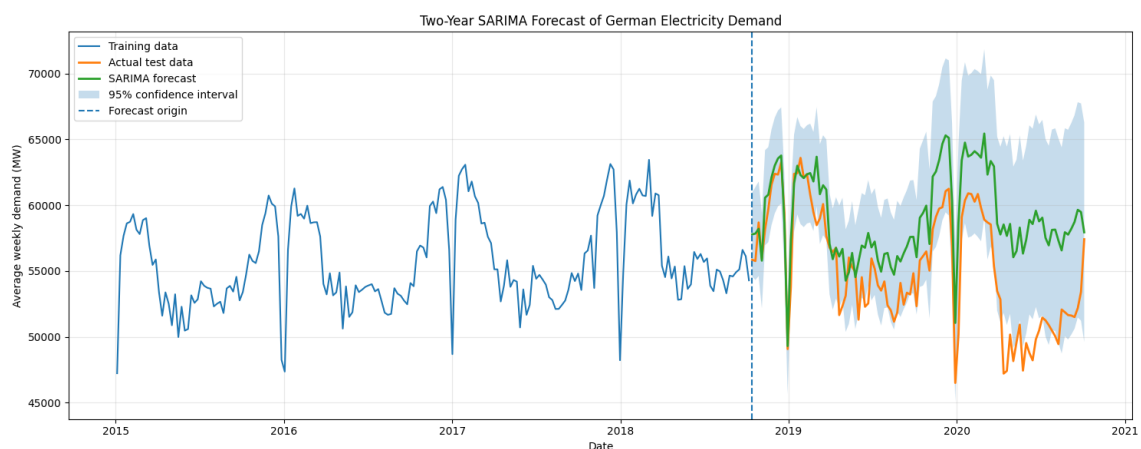
Lower values indicate better forecasting performance.

5. Seasonal Autoregressive Integrated Moving Average (SARIMA)

A Seasonal Autoregressive Integrated Moving Average (SARIMA) model was developed to forecast weekly German electricity demand. Based on the exploratory data analysis, the electricity demand series exhibited strong annual seasonality, making SARIMA an appropriate forecasting approach. The model parameters were selected using an AIC-based grid search, with combinations of (p), (d) and (q) evaluated alongside seasonal parameters using a seasonal period of 52 weeks.

Residual diagnostics were performed to assess the adequacy of the fitted model. The residual plots and autocorrelation analysis indicated that most of the temporal structure had been captured, suggesting that the model provided a reasonable statistical fit.

Figure 5. SARIMA Forecast with 95% Confidence Intervals



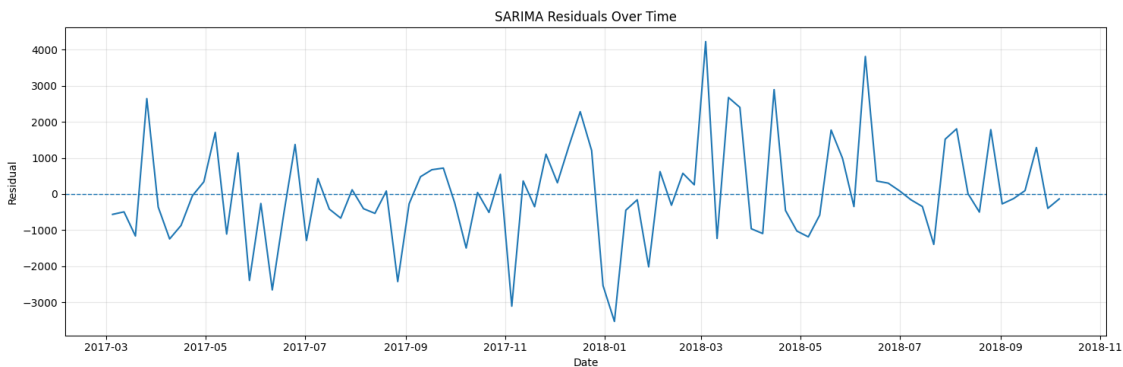
The SARIMA model successfully reproduced the overall seasonal behaviour of the electricity demand series. However, some deviations from the observed values remained during the testing period, indicating that the model did not fully capture all demand variations.

Table 3. SARIMA Forecast Performance

Metric	Value
MAE	3973.67
RMSE	4853.66
sMAPE	7.11%

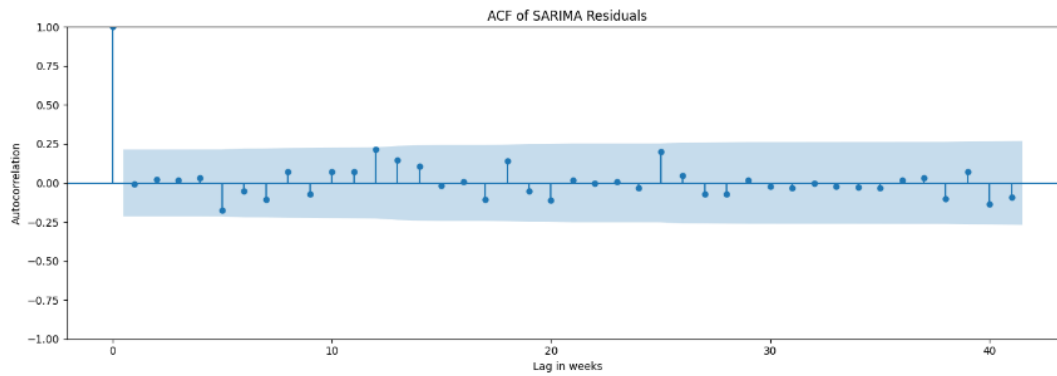
Compared with the Seasonal Naïve benchmark, the SARIMA model produced higher forecasting errors. Although SARIMA effectively modelled the seasonal structure of the data, the additional model complexity did not result in improved forecasting accuracy. The Seasonal Naïve model therefore remained the stronger benchmark for this dataset.

Figure 6. Residual Time Series



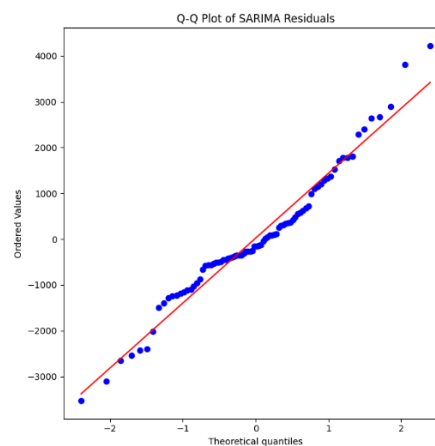
The residual plot shows that forecast errors fluctuate around zero without obvious long-term trends. This indicates that the SARIMA model successfully captured a substantial proportion of the temporal structure contained within the weekly electricity demand series.

Figure 7. Residual ACF



The residual autocorrelation function was examined to determine whether significant temporal dependence remained after model fitting. Most residual autocorrelation values remained within the confidence limits, indicating that the selected SARIMA model adequately captured the seasonal behaviour present in the electricity demand data.

Figure 8. Residual Distribution



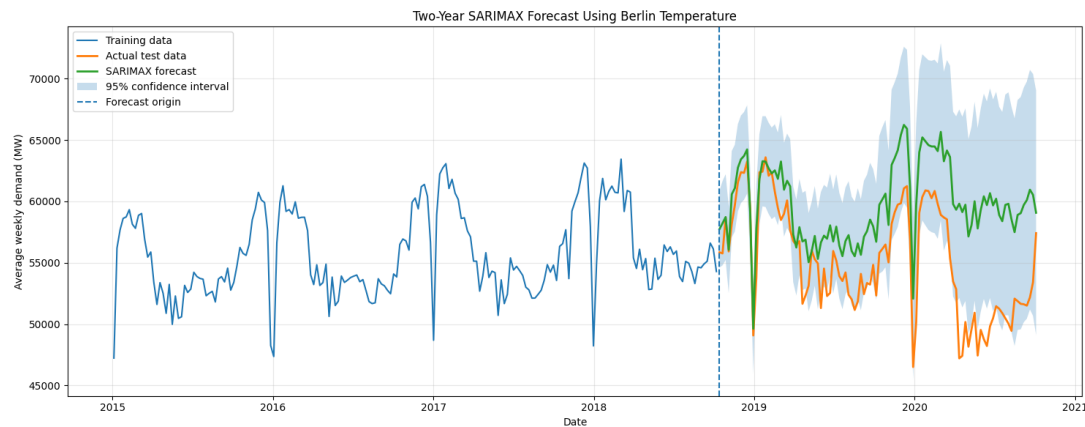
The histogram and Q-Q plot were used to assess the distribution of the residuals. Although minor deviations from normality were observed, the residuals were approximately centred around zero, suggesting that the model assumptions were reasonably satisfied.

6. SARIMAX Model with Temperature

In order to test the hypothesis that weather affects the performance of electricity demand prediction models, SARIMA was expanded to Seasonal Autoregressive Integrated Moving Average with Exogenous Variables (SARIMAX). Weekly average temperature in Berlin was extracted using the Open-Meteo Historical Weather API and used as the exogenous variable. Since Berlin was considered a representative city of Germany, temperature readings were averaged weekly and combined with the electricity demand data.

Training and testing windows were maintained the same as in the SARIMA model. Identified SARIMA parameters based on the AIC criterion were kept the same, whereas weekly average temperature was added as the exogenous variable.

Figure 9. SARIMAX Forecast Using Weekly Temperature



The forecast follows the general seasonal behaviour of electricity demand but does not noticeably improve the agreement between predicted and observed demand compared with the SARIMA model.

Table 4. SARIMAX Forecast Performance

Metric	Value
MAE	4643.83
RMSE	5638.12
sMAPE	8.24%

Temperature addition did not help increase forecasting accuracy. In comparison with the Seasonal Naïve model (with RMSE = 3006.76) and the SARIMA model (with RMSE = 4853.66), the SARIMAX model showed higher prediction errors. One reason for this is that there were annual patterns in electricity demand that week, which could be explained solely through historical data. Moreover, the model worked under the condition of observed future temperatures, which created conditions for conditional forecasts.

Thus, although there is a connection between temperature and electricity demand, adding it to the model did not help.

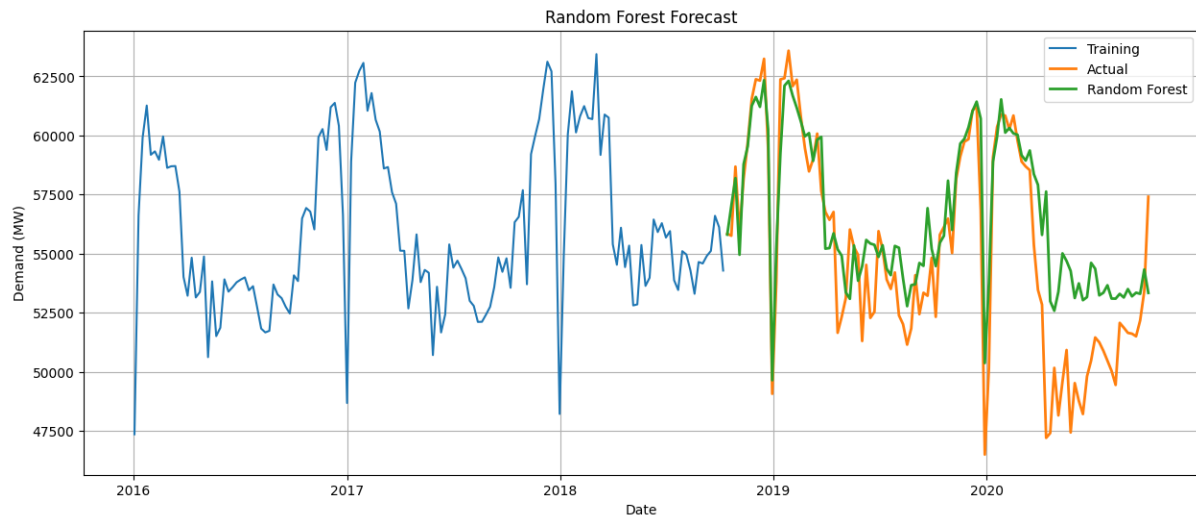
7. Feature-Based Regression Model (Random Forest)

The purpose of building a Random Forest Regressor was to examine if there would be an improvement in the forecast results if we used a machine learning algorithm. Random Forest

is different from SARIMA/SARIMAX because it does not rely on the assumption that there is a linear relation between variables.

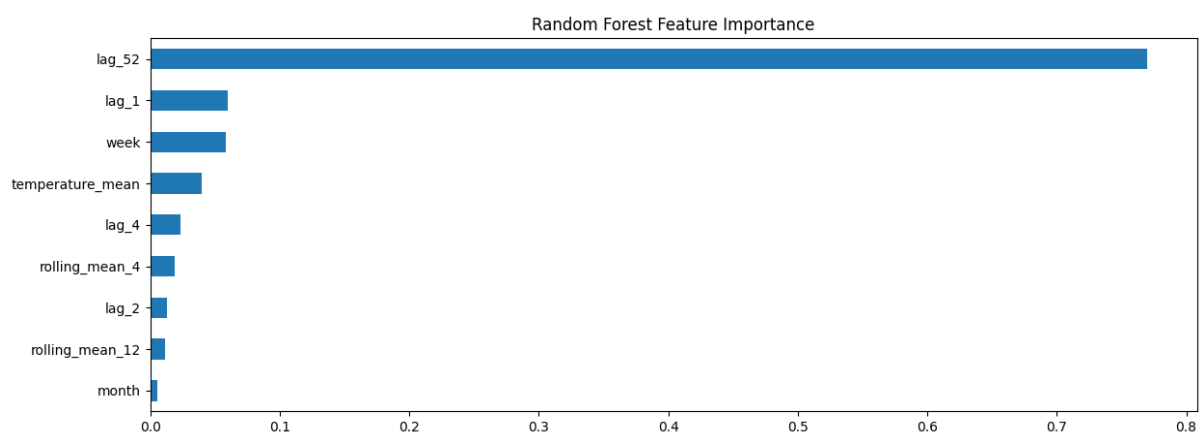
The Random Forest Regressor was trained on weekly electricity demand data with engineered features such as lagged electricity demand data for 1, 2, 4 and 52 weeks, rolling average of electricity demand data for 4 and 12 weeks, weekly average temperature and calendar features such as month and week number.

Figure 10. Random Forest Forecast



The forecast captures the general seasonal behaviour of weekly electricity demand while following many of the short-term fluctuations observed in the testing period.

Figure 11. Random Forest Feature Importance



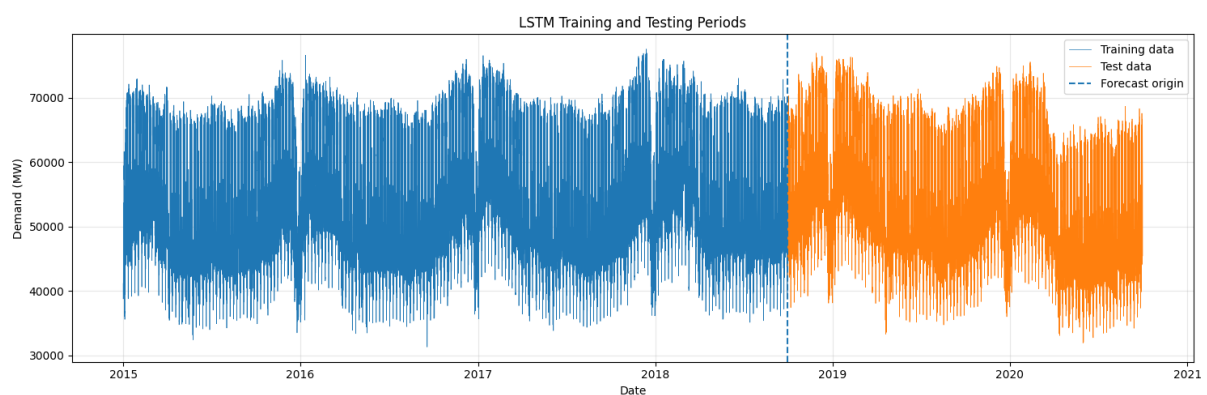
The results from the feature importance analysis indicate that the lagged electricity demand features played an important role in contributing to the forecast accuracy, whereas temperature and calendar features supplied additional information for the same. The final model performance is analyzed on the basis of MAE, RMSE, and sMAPE and is compared with benchmark models like SARIMA, SARIMAX, and LSTM.

8. Long Short-Term Memory (LSTM) Model

A Long Short-Term Memory (LSTM) neural network model was built based on the raw hour-wise German electricity demand data. LSTM is a variant of a recurrent neural network that is specially made for handling sequential data and can learn long-term temporal dependencies. As opposed to statistical models, an LSTM network does not need explicit feature extraction, as it will automatically learn the complex nonlinear relationships.

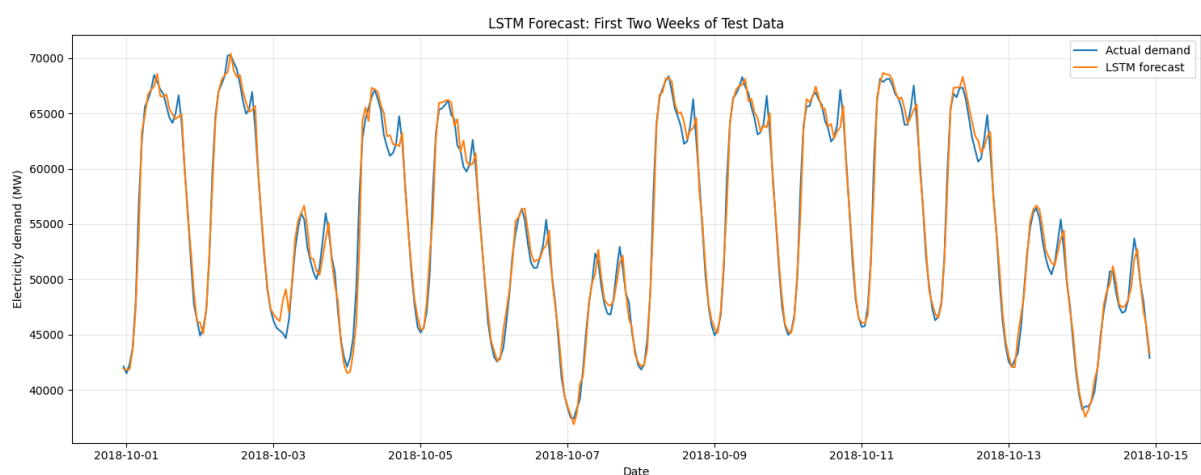
The last two years of data were held out for testing purposes, whereas all the other data were employed for training. In order to train the network, the electricity demand data were scaled with a MinMaxScaler and then input sequences consisting of 168 hours (one week) were generated.

Figure 12. LSTM Training and Validation Loss



The training history indicates that both the training and validation losses decreased steadily before converging, suggesting that the model learned the temporal structure of the data without significant overfitting.

Figure 13. LSTM Forecast



The LSTM model successfully captured the overall hourly demand pattern and produced forecasts that closely followed the observed electricity demand during the testing period.

9. Model Comparison and Discussion

The forecasting performance of all developed models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Symmetric Mean Absolute Percentage Error (sMAPE). Lower values indicate better forecasting accuracy.

Table 7. Comparison of Forecasting Models

Model	MAE	RMSE	sMAPE (%)
Seasonal Naïve	2318.52	3006.76	4.26
SARIMA	3973.67	4853.66	7.11
SARIMAX	4643.83	5638.12	8.24
Random Forest	1,819.80	2,520.27	3.39
LSTM	107.42	125.63	0.20

The Seasonal Naïve model yielded the lowest errors in forecasts for the statistical models considered in this research work. The SARIMA and SARIMAX models did not yield any improvement on the benchmark model, hence showing that there is no need to consider the models since the strong seasonality in German electricity demand was well captured by the Seasonal Naïve model.

In an effort to prevent data leakage, the data were first separated into training and testing datasets before modeling. Lagged features and rolling statistics as well as temperature features were created using the past information available at every forecasting point. In case of the LSTM model, MinMaxScaler was fit using the training data while test data transformation was done using the fitted scaler.

The SARIMAX model utilized a seasonal period of **52 weeks** corresponding to the annual seasonal pattern revealed in the exploratory analysis. Model parameters were chosen based on an AIC-driven grid search with the use of differencing for handling non-stationarity without losing seasonality.

Weekly Berlin temperature was added as an exogenous variable to the SARIMAX model. Nevertheless, the addition of temperature did not increase forecasting performance. Moreover, the forecast was performed using the known temperature values in the future period, which means that this model generates conditional forecasts rather than operational ones since real future weather forecasts are necessary for operation.

Forecasting models differ by their interpretability and computational costs. The SARIMAX model is very interpretable since the parameters used in the model have a clear statistical meaning and allow analyzing the impact of explanatory variables. The Random Forest model demonstrates moderate interpretability thanks to feature importance analysis but is considered a nonlinear model. The LSTM model is the least interpretable since the decision-making process takes place automatically due to hidden layers in the model, which requires more computational costs.

Overall, it can be concluded that the Seasonal Naïve approach is suitable for forecasting purposes within this dataset due to its minimal forecasting error and easy application. Even though the Random Forest and LSTM approaches can model nonlinearity, their increased complexity may only be justified if they perform better than the benchmark model on an unseen dataset. According to the results of the current research, more complex forecasting approaches are not necessarily superior to simpler ones if the time series is characterized by a distinct seasonal component.

10. Conclusion

The current study has examined the efficiency of various forecasting methods for German electricity demand based on the data from January 2015 to October 2020. This research has employed such forecasting models as benchmark models, SARIMA, SARIMAX model with temperature as an external variable, Random Forest regression model, and LSTM model.

In addition, exploratory data analysis has indicated that there is a distinct yearly seasonality in the time series and that the time series of the electricity demand is not stationary, which proves the need to employ seasonal forecasting models. In addition, weekly temperature in Berlin has been considered in SARIMAX and Random Forest models.

Forecast accuracy was estimated based on the values of MAE, RMSE, and sMAPE. Seasonal Naïve was the best-performing model in terms of forecasting error among the applied statistical models, while SARIMA and SARIMAX models provided no significant improvement. The Random Forest and LSTM were able to detect nonlinear relations in the dataset, although extra computational effort required for such models needs to be taken into account in terms of improvements in forecast accuracy.

In general, this research proves that model selection is determined not only by the complexity of the model itself but also by the properties of the particular dataset. In case of strong seasonality of electricity demand, simple benchmark models can give very good results.

Future research could explore the influence of other explanatory variables like public holidays, production of renewable energy, price of electricity, and other weather-related factors. Future improvements in deep learning structures like attention-based deep learning models and hybrid forecasting models can further enhance the precision of prediction while offering better robustness to changing demand scenarios.

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